Firm-Pair Merger Prediction in the Technology Sector

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Introduction

Research Question: Can firms' patent portfolios be used as a predictor of M&A pairs in the Technology Sector?

Project Overview

- M&A is a driver of growth that allows larger firms with low R&D to acquire new technology
- Building off Jan's 2014 paper to include a new feature
  - Redefining the "technological overlap" variable with one that has more of an economic interpretation
  - Instead of a "similarity score," we are measuring overlap in the probability distribution of entire patent portfolios
  - We are moving away from "how similar are two patent portfolios" to "in what way are patent portfolios interacting in the technology space?"

Presentation Outline

Time Frame Selection

Training Data & Latent Value

Vectorize Patents

Use PatentSBERTa to create 50D vectors

Firm Patent Portfolios

50D Gaussian Mixture Models

Comparing Patent Portfolios

Bhattacharyya Coefficient & Directional Technology Complementarity

Extensions

Replicating Patent Portfolios

Why 2020? | Training on Crisis Data Surfaces Latent Value

Strategic Justification

Quiet data = hidden signal

- 2020's suppressed market conditions reduce noise and force the model to identify companies with genuine latent value, not just momentum riders. A model that works in a downturn works anywhere.

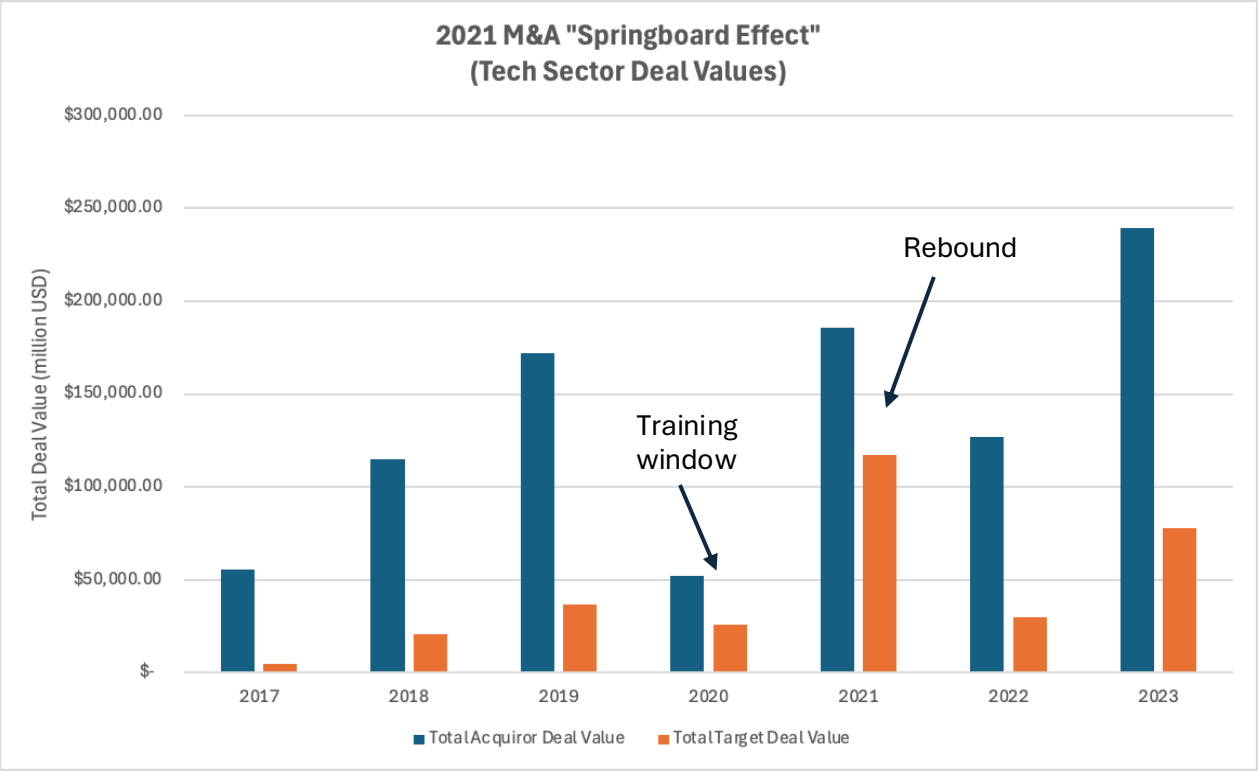
Macro-Economic Relevance

- The 2020 → 2021 recession-to-recovery arc mirrors the exact dynamic most current M&A models are built to predict. Training here gives the model real-world stress-test validity.

Data Integrity Advantage

- Conservative 2020 financials offer cleaner, less speculative inputs. Fewer inflated multiples, more grounded fundamentals that transfer to future predictions.

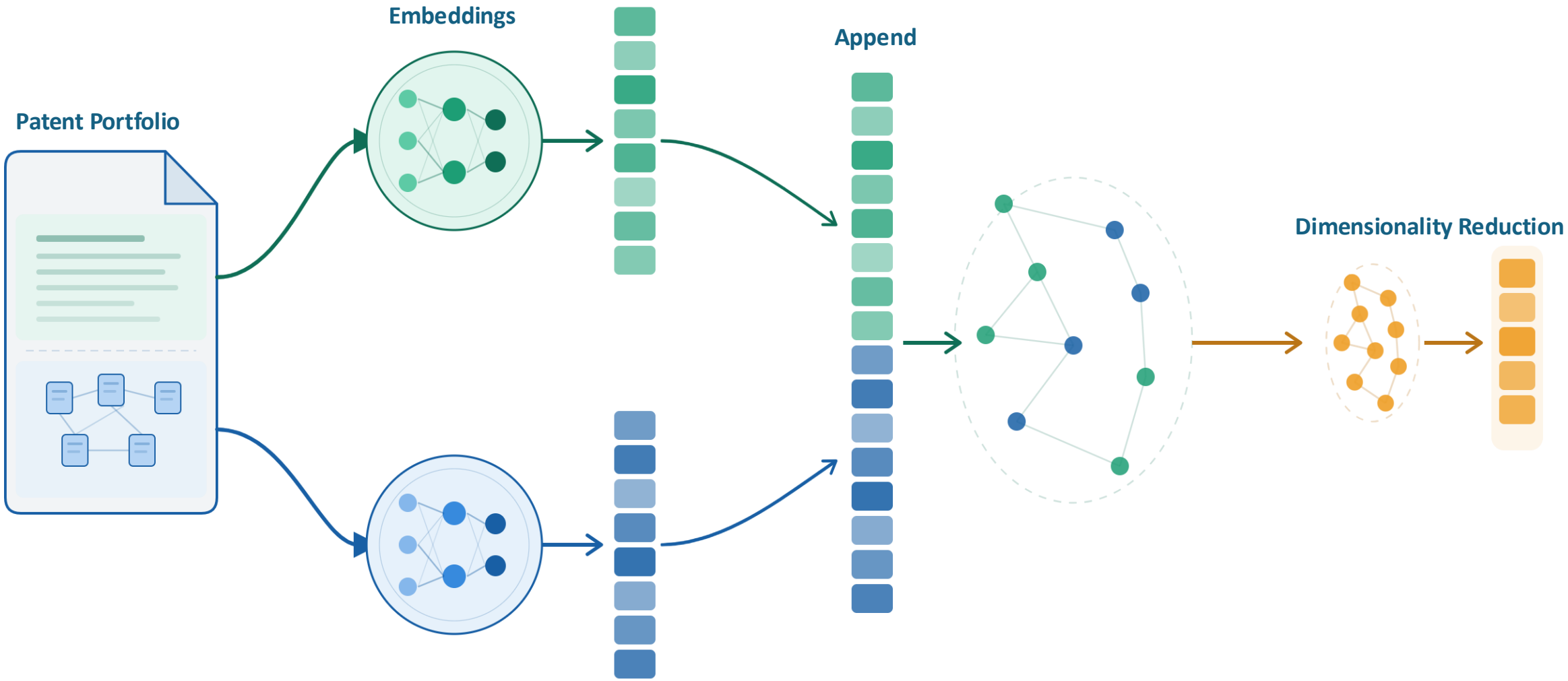
2020 vs. 2021 Value Gap



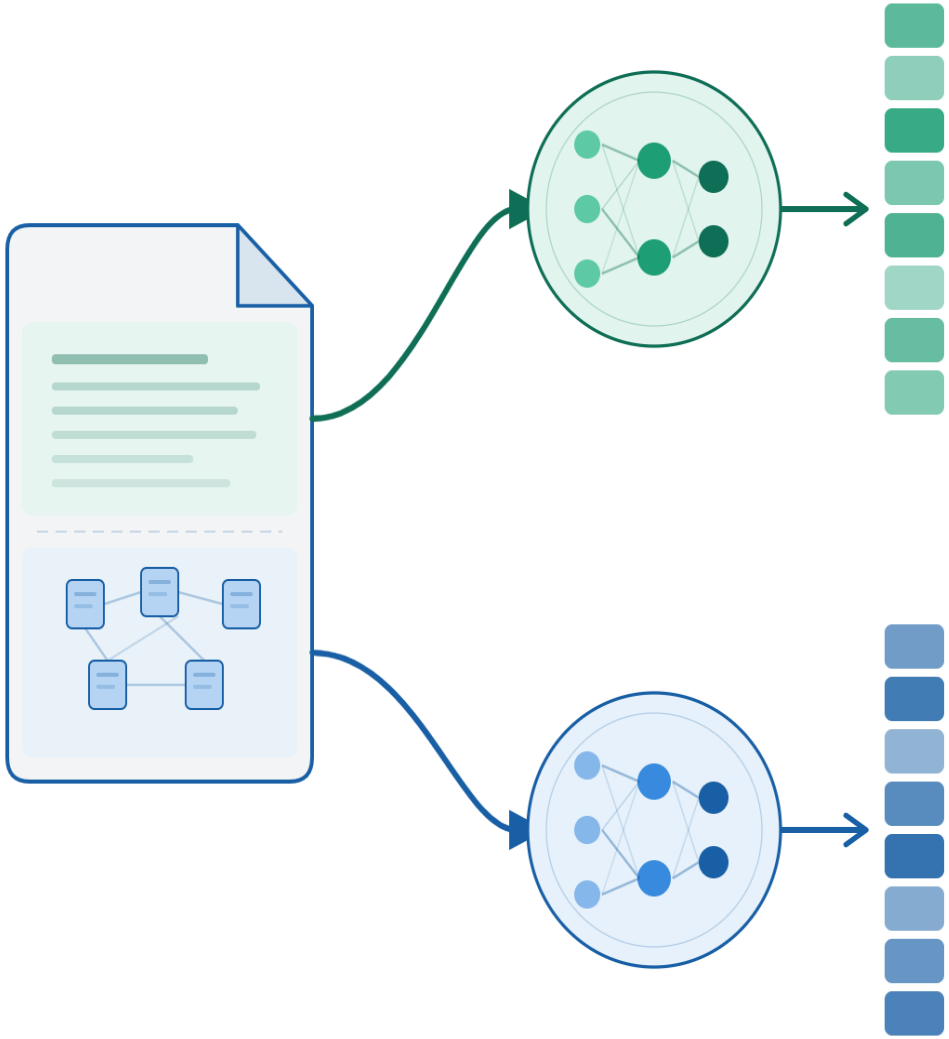
Total Target Deal Value	Total Acquiror Deal Value
$25.8B → $116.9B	$52.0B → $185.58B
+ 352% *YoY Rebound*	+ 257% *YoY Rebound*

Vectorize Patents | PatentSBERTa: From Patents to Contextual Embeddings

Pipeline Overview



Steps 1 & 2 | Patent Chunking, Parsing and Embedding



Representing Patents from Two Perspectives

1. Abstract and Title

Due to the 512 token limit of PatentSBERTa, we first embed the title and abstract to capture a general description of the invention. This encodes patent specific information in isolation of the research base this patent builds upon.

+

2. Citation Authors and Titles

Chen (2017) established that a patent citations capture knowledge base linkage. This is intuitive: patents are generally more relevant to their citations than other patents. We capture this by separately embedding citations with PatentSBERTa.

Steps 3 & 4 | Preparing for Clustering

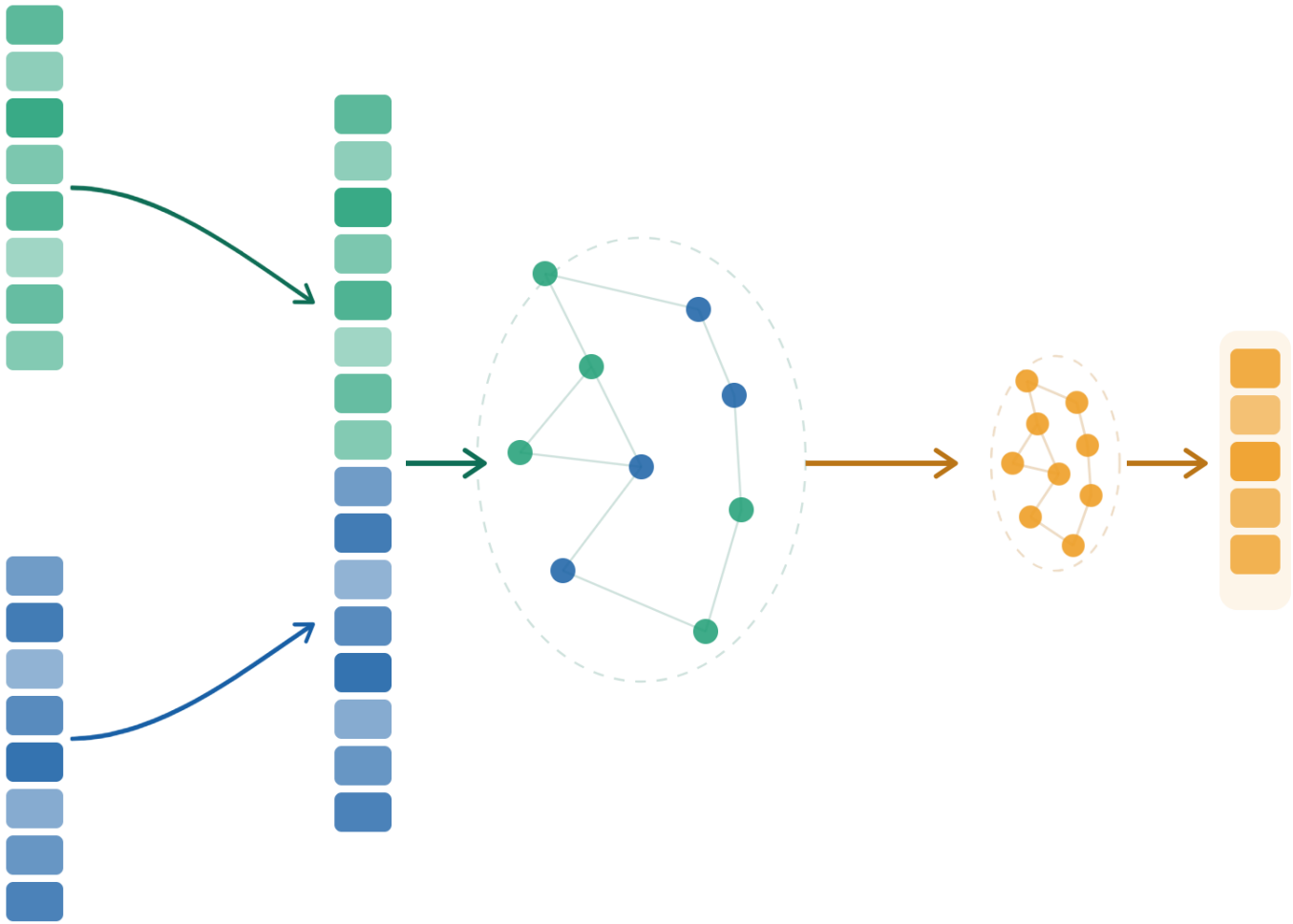
Aggregating Embeddings and Reducing Dimensionality

3. Appending Vectors

To effectively capture both dimensions of patent information (the abstract as the content of the patent, the citations as the knowledge base foundation) we concatenate the two vectors into a single "Patent Vector". This technique mirrors methods used in previous patent research by Choi et al. (2019).

4. Dimensionality Reduction

Appending these vectors comes at the cost of having a 1536 dimensional representation, which would suffer from the curse of dimensionality if used for downstream clustering. As such, we apply UMAP to map the vector into a lower dimension subspace that preserves distances between points (50D).

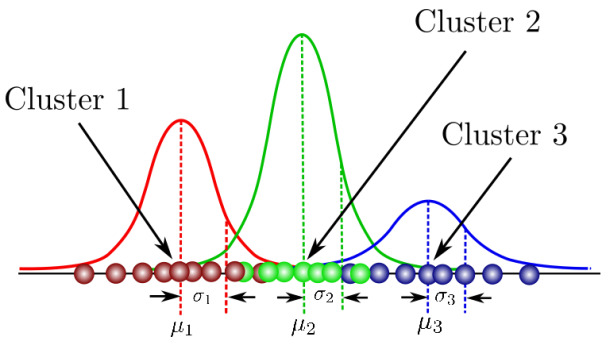


Patent Portfolio Representation | Gaussian Mixture Models

Gaussian Mixture Model

- Model set of 50D patent vectors as K (number of clusters) Gaussian probability distributions
  - Each patent has a probability of belonging to each Gaussian distribution (cluster)
- Assign weights (mixing coefficients) to clusters via expectation-maximization
  - Weights can be interpreted as "technology" and represent the patent share (by probability) of the technology within the firm

→ Gaussian Mixture Model



Determining K (Number of Clusters)

1. Brute Force Loop

Run the model for multiple clusters (multiple value of K). Analyze the Bayesian Information Criterion (BIC) to determine the fewest clusters that still explain the data.

Or

2. Bayesian Gaussian Mixture Model

Set a maximum K and a weight concentration prior. Given this prior, the clusters are automatically pruned: clusters with high frequency are given larger weight and clusters with low frequency are given little / no weight.

Other?

Ref: Dempster et al. (1977); Blei & Jordan (2006)

Patent Portfolio Representation| Assumptions & Pros/Cons

Assumptions

Patents are Related

- Assumes patents can be organized into clusters

Ellipsoid Clusters

- Gaussian shape (high dimension ellipsoid) is enforced on the distribution

Linear Superposition

- A firm's patent portfolio is the weighted sum of individual Gaussians

Correct Prior

- If using Bayesian GMM, prior needs to reflect nature of the M&A market

Pros	Cons
Soft Assignment to Clusters	Need to Determine K (Clusters) // Hyperparameter Sensitivity (Bayesian)
Accounts for Asymmetric Scale	Static Model
Automatic Pruning (Bayesian)	High Dimensionality
Computationally Efficient Comparisons	Rare Technologies?

Bhattacharyya Coefficient | Measuring GM Overlap Between Two Firms

Bhattacharyya Coefficient

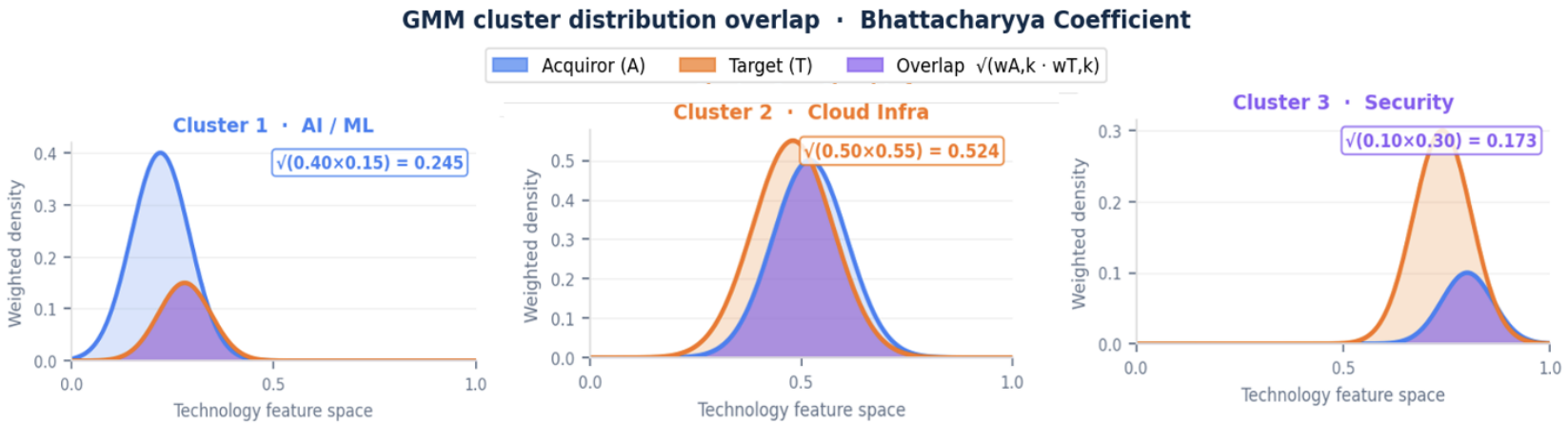
$$BC = \Sigma \sqrt{(w_{A,k} \cdot w_{T,k})}$$

Where:

- k = a specific technology cluster within GMM
- $w_{A,k}$ = Acquiror's weight (probability) in cluster k
- $w_{T,k}$= Target's weight (probability) in cluster k

Application using GMM cluster

- **Organic Technology Mapping**: Instead of static categories, use GMM to organically group patents/tech into distinct clusters based on their underlying features.
- **Company Portfolios as Weights:** Every company's technology portfolio is represented as a probability distribution across these GMM clusters (e.g., Company A is 40% Cluster 1, 50% Cluster 2, 10% Cluster 3)
- **Calculating the Overlap:** Compute BC of Acquiror and Target by aggregating the shared weights across every individual cluster.



Key Insight: Calculated BC score is fed directly into our model as a primary feature to represent strategic fit, redefining the "technology overlap" featured in Jan's 2014 paper

Directional Technology Complementarity | Asymmetric Portfolio Comparison

Motivation

- **Beyond Bhattacharyya:** BC measures overlap/similarity; we want a measure of how technologies interact
- **Core Question:** Does one firm's portfolio fill gaps in another's?
- **Literature:** Bloom, Schankerman & Van Reenen (2013) establish that firms interact through asymmetric spillover channels - motivating a directional, not symmetric, measure.

The Measure

- **GMM Representation:** Each firm is a probability distribution over GMM clusters with mixture weights p_{ak}
- **Gap term:** $(1 - p_{ak})$ captures Firm A's underexposure in cluster k; p_{2k} captures Firm B's strength there
- **Formula:** $\text{Comp}(A \leftarrow B) = \Sigma_k (1 - p_{ak}) \cdot p_{2k}$; intuitively measures how well B's strengths fill A's gaps

Interpretation vs. Bhattacharyya

BC → Overlap

Similarity / substitutability between portfolios

Comp → Gap-filling

Synergy / capability-acquisition potential

Key Properties

Directional

$\text{Comp}(A \leftarrow B) \neq \text{Comp}(B \leftarrow A)$; distinguishes acquirer vs. target role

Merger Type Classifier

High BC + low Comp → consolidation; low BC + high Comp → capability acquisition.

Extension | Synthetic Portfolio Matching

The Idea

We construct a synthetic post-merger distribution p_{A+B} and compare it to existing firms — enabling both forward validation (does A+B ≈ C?) and inverse target identification (find B such that A+B ≈ C).

Use Cases

Forward Problem

A+B ≈ C? Does acquiring B move A toward C's technological position?

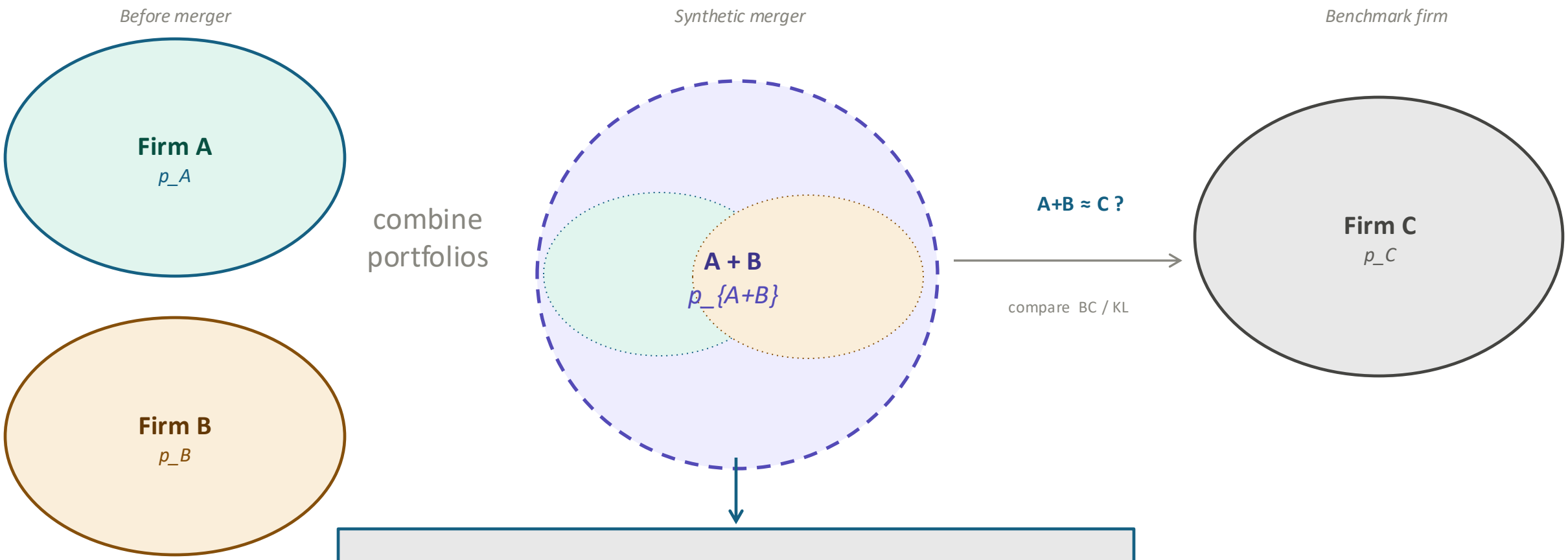
Inverse Problem

Find x s.t. A+x ≈ C, then match x to real firms in the market

Target Identification

Solve for x, then rank all firms by how closely they match - outputs a ranked acquisition shortlist

Extension | Forward Problem



If A+B ≈ C

Acquiring B repositions Firm A toward C's technological position

→ B is a strategic acquisition target for A

Extension | Inverse Problem

Known firms

Firm A
p_A

+

X
p_B

=

Firm C
p_C

solve for
x = p_C − p_A

Derived gap distribution

X
missing distribution
p_C − p_A (renorm.)

match to real firms →

Ranked targets

#1	**Firm B★**	*p_B ≈ x*	**sim: 0.89**
#2	Firm B_1		sim: 0.71
#3	Firm B_2		sim: 0.58
#4	Firm B_3		sim: 0.44

Output: ranked acquisition targets
Find B such that p_B ≈ x = p_C − p_A
→ B★ closes the gap between Firm A and Firm C

Thank you!

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Appendix

Multivariate Bhattacharyya Coefficient

m: mean matrix (1 x 50)
R: covariance matrix (50 x 50)

Multivariate Gaussian Distributions

If $p_i(x) = N(m_i, R_i)$, then

$$B = \frac{1}{8}(m_1 - m_2)'R^{-1}(m_1 - m_2) + \frac{1}{2}\ln\left(\frac{\det R}{\sqrt{\det R_1 \cdot \det R_2}}\right) \quad (61)$$

where

$$2R = R_1 + R_2. \quad (62)$$

Ref: Kailath (1967)